# Appendix

## List of Logic Statements

### 1. True Propositions

**Statement 1**. Gödel's first incompleteness theorem is applicable to Robinson's arithmetic.

**Statement 2**. De Morgan's theorems establish equivalence relations between boolean operators of conjunction and disjunction.

**Statement 3**. The alphabet of a first-order language may include symbols for individual constants and for logical connectives.

**Statement 4**. In the standard model of Peano arithmetic, the reference set for variables is the set of natural numbers.

**Statement 5**. The definable subsets of a first-order structure form a Boolean algebra with respect to the usual set operations.

**Statement 6**. In Robinson's arithmetic, it is stated that the 'successor' function is an injective function.

**Statement 7**. In the set of natural numbers, there are arithmetic subsets that are not recursively enumerable.

**Statement 8**. Zermelo-Fraenkel set theory allows a set to contain no elements.

### 2. True Inferences

**Statement 9**. If a language $L$ from a finite alphabet and its complement are semidecidable, then $L$ is a decidable language.

**Statement 10**. If two structures of the same countable first-order language are partially isomorphic, then they are elementarily equivalent.

**Statement 11**. If a first-order logic theory is categorical in the power of the continuum, then it is omega-stable.

**Statement 12**. A theory has a closed tree if and only if there is a finite closed subset of it.

**Statement 13**. If $D$ is a proof in Gentzen's system, in the same system there exists $D'$ of the same sequent that contains no cuts.

**Statement 14**. A formula in the language of ZFC is provable in NGB if and only if it is provable in ZFC.

**Statement 15**. A set $X$ of natural numbers is recursively enumerable if and only if it is the domain of a recursive partial function.

**Statement 16**. If there exists a realization that makes all formulas of a first-order theory true, then it is satisfiable.

### 3. False Propositions

**Statement 17**. The set of partial recursive functions from the set N of natural numbers to N itself has the power of a continuum.

**Statement 18**. The halting problem for the universal Turing machine is solvable.

**Statement 19**. In a Boolean algebra, a proper filter with the property of finite intersection is an ultrafilter.

**Statement 20**. The satisfiability problem for a first-order predicate logic language is always decidable.

**Statement 21**. The elimination of the negation symbol is one of the axioms of natural deduction.

**Statement 22**. The SLD-resolution calculus for Horn clauses, with the depth-first strategy for generating the search tree, is strongly complete.

**Statement 23**. For every set $X$, the cardinality of $X$ is greater than the cardinality of the power set of $X$.

**Statement 24**. Robinson's arithmetic, usually denoted by $Q$, is a complete theory.

### 4. False Inferences

**Statement 25**. A theory in first-order predicate logic that is categorical in some infinite cardinality is categorical in every infinite cardinality.

**Statement 26**. There is only one binary connective that alone constitutes a complete basis of connectives.

**Statement 27**. The class of well-ordered sets can be axiomatized in first-order logic, but only through infinite statements.

**Statement 28**. A theory in a first-order logic language, if strongly minimal, is also o-minimal.

**Statement 29**. The problem of deciding, for every natural number $x$, whether $x$ is in the image of phi $x$ zero, is recursively unsolvable if $x$ zero is a natural number.

**Statement 30**. If $T$ is a consistent theory in a first-order language $L$, then there exists and is unique a theory $T'$ in the same language extending $T$.

**Statement 31**. If $f$ is a function from the set $N$ of natural numbers to $N$ itself computable according to Turing, then $f$ is primitively recursive.

**Statement 32**. Assuming ZFC, Martin's Axiom implies the Continuum Hypothesis.

### 5. Meaningless Propositions

**Statement 33**. Elementary Peano arithmetic satisfies the non-logical constant of natural deduction.

**Statement 34**. Every consistent theory in a first-order language $L$ intersects Martin's axiom.

**Statement 35**. In a Boolean algebra, the problem of deciding computable functions in the set of natural numbers $N$ is reversible.

**Statement 36**. The power of the continuum is modellable according to the product of two cardinals $h$ and $k$, both different from 0, of which at least one is infinite.

**Statement 37**. Given the cardinal numbers $h$ and $k$ greater than one, the powers of $h$ to the $k$ and $k$ to the $h$ are semidecidable.

**Statement 38**. The addition of the negation symbol in a Martin's axiom results in the removal of the ultrafilter.

**Statement 39**. All propositional formulas in a decidable space are recursive and primitive.

**Statement 40**. Among the satisfiability functions of the set $N$ of positive statements, there exist infinitely many recursive proportions.

### 6. Meaningless Inferences

**Statement 41**. First-order logic is enumerable only if partially recursive.

**Statement 42**. The Goldbach conjecture is infinite and deducible only in a propositional space.

**Statement 43**. The definable subsets of a first-order structure satisfy the Lowenheim-Skolem property only if recursively enumerable.

**Statement 44**. A set $X$ of natural numbers is complete if and only if it satisfies the Goldbach conjecture.

**Statement 45**. If $A$ is a first-order structure and $A$ prime is an equivalent substructure, the compactness principle is not satisfied.

**Statement 46**. There exist infinitely many first-order cardinal numbers for which the respective class of well-ordered sets is undecidable.

**Statement 47**. If a language $L$ is decidable in an alphabet $A$, every non-logical constant appearing in $L$ is semidecidable in $A$.

**Statement 48**. If a proposition is deducible according to the Lebesgue axiom, then it is negatable in its space of existence.

## List of Non-logic Statements

### 7. True Propositions

**Statement 49**. In all ancient cultures that emerged in the Mediterranean, bulls were worshipped as deities.

**Statement 50**. In Ancient Greece, a citizen who could not pay his debts became a slave.

**Statement 51**. Starting from the 1970s, the General Income Tax was replaced as a tax by the Value Added Tax.

**Statement 52**. Having contracted a previous marriage bond is one of the impediments to marriage provided for by Canon Law

**Statement 53**. Besides Vatican City, the Principality of Monaco is the smallest country in the world.

**Statement 54**. Early versions of the concepts of 'robot' and 'avatar' were already present in Greek mythology.

**Statement 55**. The most recent Nobel Prize in Physiology awarded to an Italian woman was given to Rita Levi-Montalcini.

**Statement 56**. Human brain is divided into 4 lobes.

### 8. True Inferences

**Statement 57**. Since the last constitutional referendum resulted in a NO vote, the CNEL was not abolished.

**Statement 58**. If Italo Calvino had been born in the second half of the 20th century, he would not have been able to commit himself during the Resistance.

**Statement 59**. Aosta is the only province in its region, which also makes it the capital.

**Statement 60**. Since Italy is a peninsula of the Adriatic Sea, it is not washed by the waters of the Pacific Ocean.

**Statement 61**. It is the presence of both a Parliament and a monarch that makes Spain a parliamentary monarchy.

**Statement 62**. Being the product of the modification of the ovary following fertilization, the tomato is a fruit.

**Statement 63**. Since it evolutionarily descends from monkeys, the tamarin is a primate.

**Statement 64**. Being a member of the *Phylum Chordata*, the barn owl consequently belongs to the domain of Eukaryotes.

### 9. False Propositions

**Statement 65**. The Milan subway was built before the Paris subway.

**Statement 66**. All borders in Europe, except for Yugoslavia, were created at the end of World War II.

**Statement 67**. The director Pier Paolo Pasolini joined the Italian Communist Party five years after the fall of the Berlin Wall.

**Statement 68**. The end of the Council of Trent coincides with the fall of the Western Roman Empire.

**Statement 69**. Cetaceans have a respiratory system that allows them to breathe beneath the water's surface.

**Statement 70**. In every society in the world, the market is considered an essential and foundational institution.

**Statement 71**. The Nuclear Non-Proliferation Treaty prohibits the countries involved from developing technologies for the peaceful use of nuclear energy.

**Statement 72**. The alternating high and low tides on Earth are determined by the position of Uranus relative to Earth.

### 10. False Inferences

**Statement 73**. Since Garibaldi was a general, the 1860 Expedition of the Thousand was led by a simple soldier.

**Statement 74**. Since the Gulf Stream reaches England, its winters are extremely cold.

**Statement 75**. It is thanks to the merger between the car manufacturers Fiat and Peugeot that in 2014 the Italo-American group FCA was born.

**Statement 76**. The papacy was moved to Avignon for about 70 years, so it was never outside Italian territory.

**Statement 77**. If the European Union had not introduced the single currency, the Pound would still be in force in Italy.

**Statement 78**. Being a grandfather, the Duke of Windsor never had children.

**Statement 79**. Since this individual was permanently hired, they transitioned to unemployment.

**Statement 80**. Since female ostriches lay eggs, they are to be considered as examples of mammals.

### 11. Meaningless Propositions

**Statement 81**. The fully green potato flag was guillotined at the end of the Council of Trent.

**Statement 82**. Institutionalized market drinks one of the first concepts of western roman avatars known to us.

**Statement 83**. All beans in debt after the fall of the Western Roman Empire have a scientific background.

**Statement 84**. Ancient Greece mythology is the smallest alcoholic derivation of the VAT tax.

**Statement 85**. Most of the robotic bulls of the Mediterranean have never met Yugoslavia.

**Statement 86**. An ancient poet is a predominantly green treatise on the subway of the involved countries.

**Statement 87**. The Molise region coincides with the recourse to Greek energy in the nuclear mythology of the Communist Party.

**Statement 88**. The alternating high and low tides on Earth was abolished by the construction of the barn owl in 1905.

### 12. Meaningless Inferences

**Statement 89**. Since Garibaldi did not introduce the common currency, the Duke of Windsor developed a visual system.

**Statement 90**. The papacy introduced the depth of the ocean, therefore in Italy the monarch was born.

**Statement 91**. Aosta is the successor of Fiat, then in 2014 the European Union developed children.

**Statement 92**. If the last constitutional referendum had been directed by a winter, the tamarin would have been outside Italian territory.

**Statement 93**. Without the simultaneous presence of Italo Calvino and Peugeot, the common currency would not have been achieved.

**Statement 94**. Having developed from the waters of the CNEL, the Italian territory was moved to the province of the 20th century.

**Statement 95**. Since it descends from Avignon, Spain is the result of the frozen evolution.

**Statement 96**. Since female ostriches lay common currencies, then Yugoslavia is determined by the position of Uranus.